

Discussion 12: Maxwell's equations and relativity – Solutions

Names:

Section:

This week we will start to see the connection between electrodynamics and relativity. We'll derive the last of Maxwell's equations in differential form, and then see how relativity *requires* that magnetic and electric fields are related by Lorentz transformations!

Part 1: Ampère's Law in differential form

In PHYS 212, you saw that the example of a Gaussian surface enclosing the gap between the plates of a capacitor implies there must be a displacement current correction to Ampère's Law:

$$\oint_C \mathbf{B} \cdot d\mathbf{r} = \mu_0 I_{\text{encl}} + \mu_0 \epsilon_0 \frac{d}{dt} \int_A \mathbf{E} \cdot d\mathbf{S}, \quad (1)$$

where C is the closed curve which bounds the surface A (in other words, $C = \partial A$).

- 1.1. Use Stokes' Theorem to write the left-hand side of (1) as a surface integral of $\nabla \times \mathbf{B}$.

$$\oint_C \mathbf{B} \cdot d\mathbf{r} = \int_A (\nabla \times \mathbf{B}) \cdot d\mathbf{S}$$

- 1.2. Define the enclosed current in terms of the current density $\mathbf{J}$ as $I_{\text{encl}} \equiv \int_A \mathbf{J} \cdot d\mathbf{S}$, where A is the same surface as appears in (1). Use this to write the right-hand side of (1) as one combined surface integral over A .

Since $\mu_0 I_{\text{encl}} = \mu_0 \int_A \mathbf{J} \cdot d\mathbf{S}$, we can combine the two terms into the same surface integral by linearity of integration:

$$\mu_0 I_{\text{encl}} + \mu_0 \epsilon_0 \frac{d}{dt} \int_A \mathbf{E} \cdot d\mathbf{S} = \int_A \left(\mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \cdot d\mathbf{S}.$$

Note that we have turned the time derivative into a partial derivative by pulling it inside the integral sign, since the integrand $\mathbf{E}$ does depend on spatial coordinates.

- 1.3. Argue that since A is arbitrary (and can be taken to be arbitrarily small), the integrands on both sides must be equal, and thus obtain the differential form of Ampère's Law as given in lecture.

Taking A to be arbitrarily small lets us pull the integrand out of the integral, and thus the integrands must be equal. Alternatively, we can move the RHS to the LHS and argue that since the surface integral of RHS – LHS is zero for any arbitrary surface A , the integrand must be zero. Either case yields

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Part 2: Lorentz transformation of a current-carrying wire

Consider the following situation: A *neutral* thin wire consists of a string of positive charges moving to the right at speed v , and a string of negative charges moving to the left at speed v as well. Assume there are enough charges that we can regard both the positive and negative string as continuous line charges, with uniform charge per unit length λ and $-\lambda$, respectively. The wire is neutral because the total charge density is $\lambda + (-\lambda) = 0$. In addition, there is a point charge q at a distance s from the wire, moving to the right with speed w . Let's say that all this takes place in some inertial reference frame S . For this part only, we will restore all the c 's in order to work with the usual SI forms of Maxwell's equations.

- 2.1. Draw a diagram of the situation described. Show that the total current in the wire in S is $I = 2\lambda v$, moving to the right. (This is very simple).

The current is defined as charge per unit time, which has the same units as λv . The positive charges are moving to the right, so there is current λv to the right, but from the negative charges moving to the left, there is also current $-\lambda(-v) = \lambda v$. Adding these gives $I = 2\lambda v$.

- 2.2. What is the force on the charge q (magnitude *and* direction), in the frame S ? Is it an electric or magnetic force? Describe the path of the charge under the influence of this force.

The wire sources a magnetic field $\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$ in the $\hat{\phi}$ direction, where we are working in cylindrical coordinates with the wire along the z -axis. (By the right-hand rule, if you put your right thumb along the direction of the wire, the B -field curls around the wire in the same direction as the fingers on your right hand.) The magnetic force on the charge is $\mathbf{F} = q\mathbf{w} \times \mathbf{B} = qBw(\hat{\mathbf{z}} \times \hat{\phi}) = -qBw\hat{\rho}$. So if $q > 0$ and $\lambda > 0$, the force on the charge is toward the wire.

- 2.3. Now consider a reference frame S' , moving to the right with speed w . Note that in S' , the charge q is *not moving*, since the reference frame moves along with the charge. In S' , is there a magnetic force on q ?

If the charge is not moving, then the Lorentz force $q\mathbf{v} \times \mathbf{B}$ must be zero!

We seem to have a conundrum! S and S' are both inertial frames, so by the first postulate of relativity, we *must* observe the same behavior of q , and there must be a force on q in S' as well. Let's figure out what it is...

- 2.4. Consider the behavior of the wire in S' . Let $v_{\pm}$ be the velocities of the positive and negative charges, respectively, in S' . Use velocity addition to show that

$$v_{\pm} = \frac{v \mp w}{1 \mp vw/c^2} \quad (2)$$

(Be careful with signs here!)

We want to relativistically add w , the velocity of S' with respect to S , to v and $-v$, which are the velocities of the positive and negative charges in S . Since the positive charges are moving in the same direction as frame S' , v_+ must have the minus sign in the numerator, since the limiting case where $v = w$ would give $v_+ = 0$. We can then obtain v_- from v_+ just by changing v to $-v$ in the velocity addition formula.

- 2.5. Since the positive and negative charges are moving in S , length contraction implies that the charge density λ in S is different from λ_0 , the charge density in the rest frame of the charges. Use length contraction to express λ_0 in terms of λ and $\gamma_v = 1/\sqrt{1 - v^2/c^2}$. Since the positive and negative charges are moving with equal speeds in S , the length contractions for positive and negative charges are equal, and the wire stays neutral.

The charge density λ_0 is the charge divided by the “proper length.” Since the proper length gets length-contracted by γ_v for charges that are moving with speed v , the charge density increases, and γ_v goes in the numerator: $\lambda = \gamma_v \lambda_0$. We can also invert this to write $\lambda_0 = \lambda/\gamma_v$.

- 2.6. However, the situation is very different in S' ; since $v_+ \neq v_-$, the length contraction is different for the positive and negative charges, and the wire actually *acquires a net charge*. Using length contraction in S' , show that the positive and negative charge densities in S' are $\lambda_{\pm} = \pm(\gamma_{\pm})\lambda_0$, where

$$\gamma_{\pm} = \frac{1}{\sqrt{1 - v_{\pm}^2/c^2}}. \quad (3)$$

We follow exactly the same logic as problem 2.5, but this time with velocities $v_{\pm}$. The charge density is always greater in a frame where the charges are moving, so $\lambda_{\pm} = \pm(\gamma_{\pm})\lambda_0$. The $\pm$ is just expressing the fact that the charges moving with speed v_+ are positive and the charges moving with speed v_- are negative.

- 2.7. Plug Eq. (2) into Eq. (3) to show that $\gamma_{\pm}$ can be expressed as

$$\gamma_{\pm} = \gamma_v \gamma_w (1 \mp vw/c^2), \quad (4)$$

where $\gamma_w = 1/\sqrt{1 - w^2/c^2}$, and thus that the *total* charge density of the wire, λ_{tot} , is

$$\lambda_{\text{tot}} = -2\lambda(vw/c^2)\gamma_w. \quad (5)$$

Notice that this vanishes when $w = 0$, which is the original description in frame S .

To avoid writing too many $\pm$ symbols, we'll just do the calculation for γ_+ , but the logic is identical for γ_- . We'll also switch to natural units, and then restore the factors of c at the end by dimensional analysis.

$$\begin{aligned} \gamma_+ &= \frac{1}{\sqrt{1 - \left(\frac{v-w}{1-vw}\right)^2}} \\ &= \frac{1}{\sqrt{1 - \frac{v^2+w^2-2vw}{1+v^2w^2-2vw}}} \\ &= \frac{1}{\sqrt{\frac{1+v^2w^2-v^2-w^2}{1+v^2w^2-2vw}}} \\ &= \frac{1}{\sqrt{\frac{(1-v^2)(1-w^2)}{(1-vw)^2}}} \\ &= \frac{1}{\sqrt{1-v^2}} \frac{1}{\sqrt{1-w^2}} (1-vw). \end{aligned}$$

We can restore the factors of c by dividing each velocity by c , yielding the desired expression. Then, to find the total charge of the wire, we have

$$\begin{aligned}\lambda_{\text{tot}} &= \lambda_+ + \lambda_- = \gamma_+ \lambda_0 - \gamma_- \lambda_0 \\ &= \gamma_v \gamma_w (1 - vw/c^2) \lambda_0 - \gamma_v \gamma_w (1 + vw/c^2) \lambda_0 \\ &= \gamma_v \gamma_w \lambda_0 (-2vw/c^2) \\ &= \gamma_w \lambda (-2vw/c^2)\end{aligned}$$

In the last step we used the result from problem 2.4.

- 2.8. Now we're back on familiar ground: we have a thin wire with a uniform line charge, and a charge at rest a distance s from that wire. What is the electric field created by the wire, in the frame S' ? What is the force F' on the charge q ?

The electric field created by a line charge λ_{tot} is $E = \frac{\lambda_{\text{tot}}}{2\pi\epsilon_0 s}$ in the $\hat{\rho}$ direction. The force on the charge q is $\mathbf{F} = q\mathbf{E}$, which (if q , v , w , and λ are all positive) is in the $-\hat{\rho}$ direction (toward the wire), since $\lambda_{\text{tot}} < 0$. Note that the distance to the wire is s in both S and S' , because there is no transverse length contraction.

- 2.9. We have now computed the force on the charge q in two different inertial frames. Use the Lorentz transformation properties of the force (careful about the direction!) to transform F' back into S . Using the fact that the constants ϵ_0 and μ_0 are related by $c^2 = 1/(\epsilon_0\mu_0)$, show that in fact, the *electric* force in S' is identical to a *magnetic* force in S due to a uniform current I :

$$F = -qw \left(\frac{\mu_0 I}{2\pi s} \right) \quad (6)$$

This shows that magnetism is implied by relativity; the postulates of relativity plus the existence of an electric force *require* a magnetic force as well!

The force in S' is perpendicular to the Lorentz boost that takes S' to S . Therefore we want to use the rule for transforming the perpendicular component of the force, which by Morin (4.25) is $\mathbf{F}_S = \mathbf{F}_{S'}/\gamma_w$. The force stays in the perpendicular direction in S . Plugging in the expression for λ_{tot} , we have

$$F_S = \frac{1}{\gamma_w} \frac{-2q\gamma_w \lambda vw/c^2}{2\pi\epsilon_0 s} = -qw \frac{2\lambda v}{2\pi s} \frac{1}{c^2\epsilon_0} = -qw \frac{I}{2\pi s} \mu_0,$$

which is exactly the magnetic force in S .